



AMI-EV

Human eye-inspired technology could revolutionise smartphone cameras!

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If you have ever taken a deep dive into the camera technology that we have around us, you would know that the basic operating principle of cameras, as we know them today, is based on the human eye. The imaging technology that we see around us draws very close to how the human eye works. However, it turns out that there are additional attributes of the human eye that could significantly improve the way we capture images. A team of researchers led by the folks at the University of Maryland's College of Computer, Mathematical, and Natural Sciences have incorporated 'microsaccades' into their latest camera mechanism, dubbed "Artificial Microsaccade-Enhanced Event Camera (AMI-EV)", which could improve our imaging technology significantly. The applications

of this technology go beyond improving smartphone image capturing or robots. These findings were published in a research article titled – Microsaccade-inspired event camera for robotics – published in Science Robotics, Vol 9, Issue 90.

Read along to find out how Artificial Microsaccade-Enhanced Event Camera (AMI-EV) could draw the outline of a clearer digital imaging future for us –

Micromovements for the win!

Let's start by understanding the human eye's attribute called microsaccades, which are the foundation for this innovation. To put it simply, microsaccades are the involuntary movements of the human eye when it is trying to fixate on a stationary subject.

A more formal definition of microsaccades can be found in the research

paper titled – Microsaccades: A Neurophysiological Analysis – published in 2009. Researchers in the paper describe these movements as: "...the largest and fastest of the fixational eye movements, which are involuntary eye movements produced during attempted visual fixation."

When the researchers at UMD began their work, their primary aim was to incorporate this into a camera module that would be able to capture images without the motion blur that we tend to see in cameras present today. "Event cameras are a relatively new technology better at tracking moving objects than traditional cameras, but today's event cameras struggle to capture sharp, blur-free images when there's a lot of motion involved," said the paper's lead author Botao He, a computer science Ph.D. student at UMD, elaborating on the limitations of the cameras that we have today.

Over the years, researchers across the world have been trying to perfect digital imaging technology by incorporating more and more elements from the human eye into their tech. We have seen several technologies like Sensor-Shift, OIS, and EIS making their way into cameras across the board, and additional tech like Gimbals, trying to solve the issue of blurry pictures and videos. No matter how much brands develop their tech, there are limitations that are holding them back from producing absolutely flawless images like the human eye. However, AMI-EV seems to be the answer to this!

How does it work?

The basic development principle was simple – add a layer of stabilisation or information collection that could then be processed to create clear and crisp images. To accomplish this, the researchers added a rotating wedge prism right in front of the aperture of an event camera to ensure that the path of light was redirected which in turn triggered events. For the uninitiated, event cameras are a specialised set of cam-